

Problem 1, Cell 6: N lines in $\mathbb{R}^d$ (partial)

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Official statement (Cell 6, open question). “Fejes Tóth’s conjecture from the Setting, for every N and every d . For N lines in $\mathbb{R}^d$, write $N = qd + s$ with $0 \leq s < d$, and let $M(N, d) = s\binom{q+1}{2} + (d-s)\binom{q}{2}$. Prove or disprove: every N lines in $\mathbb{R}^d$ satisfy $S \leq \left(\binom{N}{2} - M(N, d)\right) \cdot \pi/2$. This is the value attained by splitting the lines as evenly as possible among d mutually orthogonal directions. Settling any infinite family not already covered above counts as partial progress.”

Status: Partial. *Proved:* (i) the conjecture for $N = 6$ lines in $\mathbb{R}^3$ (Theorem 2), by averaging Cell 4/5; (ii) the exact list of what subset averaging can give from the cases proved so far. For $d \geq 3$ and $N \geq d+3$ it is the single pair $(6, 3)$, so averaging yields *no* new infinite family (Proposition 3); (iii) a structural reduction of the case $N = d+3$ for all d . At a saturated minimiser, every component of the non-orthogonality graph is settled, except components of rank ≥ 4 , nullity ≥ 3 , maximum degree 3 and cyclomatic number ≥ 2 (Theorem 6). *Not proved:* the conjecture in general, and the case $N = d+3$ for any $d \geq 4$. The first open instance is 7 lines in $\mathbb{R}^4$. We do *not* claim a new infinite family.

1 Setting and known cases

Lines are given by unit vectors $x_1, \dots, x_N \in \mathbb{R}^d$, $g_{ij} = \langle x_i, x_j \rangle$,

$$A(x) = \sum_{i < j} \arcsin |g_{ij}|, \quad S = \binom{N}{2} \frac{\pi}{2} - A.$$

So the conjecture reads $A \geq M(N, d) \pi/2$. For earlier work on the problem see [1, 2]. Writing $n_1, \dots, n_d$ for the sizes of an even split, $M(N, d) = \sum_i \binom{n_i}{2}$. Hence $M(N, d) \geq N - d$ (since $\binom{n}{2} \geq n - 1$ for integers $n \geq 0$), and $M(N, d) = \max(0, N - d)$ for $N \leq 2d$. Also A only increases when lines are added, since all terms are ≥ 0 . The following cases are proved in our earlier cells and used here.

(C1) $d = 2$, all N : $A \geq M(N, 2) \pi/2$.

(C3) $N = d + 1$, all $d \geq 1$: $A \geq \pi/2$.

(C5) $N = d + 2$, all $d \geq 1$: $A \geq \pi$. This contains C4 (five lines in $\mathbb{R}^3$, six in $\mathbb{R}^4$).

From C5 we also use the orthogonality-saturation proposition, the path lemma, the block-structure lemma and the cycle lemma (C5, Prop. 1 and Lemmas 2–4). These give a minimiser of A in which each vertex of the non-orthogonality graph J ($ij \in J$ iff $g_{ij} \neq 0$) is either isolated or has orthogonal

partners spanning $\geq d - 1$ dimensions. For $N = d + k$: every vertex of J has degree $\leq k$. Distinct components C of J span orthogonal subspaces of dimensions r_C with $\sum_C r_C \leq d$, and $A = \sum_C A_C$. A vertex of C of degree $\delta \geq 1$ forces $r_C \leq m_C - \delta$, where $m_C = |C|$. Write $\nu_C = m_C - r_C$ (the nullity of C).

2 Subset averaging

Lemma 1 (Averaging). *If every n lines in $\mathbb{R}^d$ satisfy $A \geq B\pi/2$, then every $N \geq n$ lines in $\mathbb{R}^d$ satisfy*

$$A \geq \frac{N(N-1)}{n(n-1)} B \frac{\pi}{2}.$$

Proof. Sum A over the $\binom{N}{n}$ subsets of size n . Each pair $\{i, j\}$ lies in $\binom{N-2}{n-2}$ of them, so $\binom{N-2}{n-2} A \geq \binom{N}{n} B \frac{\pi}{2}$, and $\binom{N}{n} / \binom{N-2}{n-2} = \frac{N(N-1)}{n(n-1)}$. $\square$

Theorem 2. *Any 6 lines in $\mathbb{R}^3$ satisfy $A \geq 3\pi/2 = M(6, 3)\pi/2$, i.e. $S \leq 12 \cdot \frac{\pi}{2} = 6\pi$. Equality holds for three orthogonal directions, each taken twice.*

Proof. By C4/C5, every 5 lines in $\mathbb{R}^3$ have $A \geq \pi$. There are $\binom{6}{5} = 6$ five-subsets, and each pair lies in $\binom{4}{3} = 4$ of them. So $4A \geq 6\pi$, and $A \geq \frac{3}{2}\pi$. Here $M(6, 3)$ has $q = 2$, $s = 0$, so $M = 3\binom{2}{2} = 3$. Also $\binom{6}{2} - 3 = 12$. For three orthogonal doubled directions, the three coincident pairs give $A = 3 \cdot \frac{\pi}{2}$. $\square$

Proposition 3 (Averaging gives no infinite family). *Let $d \geq 3$ and $N \geq d+3$. Averaging (Lemma 1) from any of the cases proved in C3, C5 or Theorem 2 gives $A \geq M(N, d)\pi/2$ only for $(N, d) = (6, 3)$.*

Proof. For $d \geq 3$ the available sources are $n \leq d$ ($B = 0$, useless), $n = d + 1$ ($B = 1$), $n = d + 2$ ($B = 2$), and $(6, 3)$ with $B = 3$. C1 concerns $\mathbb{R}^2$ and does not apply to lines spanning more. The averaged bound is $\frac{N(N-1)}{n(n-1)} B$. Its coefficient $B/(n(n-1))$ is $\frac{1}{d(d+1)}$, $\frac{2}{(d+1)(d+2)}$ and (for $d = 3$) $\frac{3}{30} = \frac{2}{20}$. For $d \geq 2$ the second is the largest ($2d \geq d+2$). So we must decide when

$$\beta(N, d) := \frac{2N(N-1)}{(d+1)(d+2)} \geq M(N, d).$$

Range $d + 3 \leq N \leq 2d$. Here $M = N - d$. Put $N = d + k$, $3 \leq k \leq d$. The condition is $f(k) := 2(d+k)(d+k-1) - k(d+1)(d+2) \geq 0$, and f is convex in k . We have $f(3) = (d+2)(3-d)$ and $f(d) = d(8d-4-(d+1)(d+2)) = -d(d-2)(d-3)$. For $d \geq 4$ both are negative, so $f < 0$ on $[3, d]$. For $d = 3$ only $k = 3$ occurs, and $f(3) = 0$: this is $(6, 3)$, with equality.

Range $N \geq 2d + 1$. By convexity $M(N, d) \geq d\binom{N/d}{2} = \frac{N(N-d)}{2d}$. It suffices that $4d(N-1) < (d+1)(d+2)(N-d)$, i.e. $(d^2 - d + 2)N > d(d+1)(d+2) - 4d$. The left side increases with N . At $N = 2d + 1$ the difference of the two sides is $d^3 - 4d^2 + 5d + 2 = d(d-2)^2 + d + 2 > 0$. So $\beta < M$ for all $N \geq 2d + 1$. $\square$

Remark 4. Averaging can also use sources with a different n at once. This does not help, because the averaged bound from the case (N', d) just obtained is at most the bound from the original source (the factors $\frac{N(N-1)}{n(n-1)}$ multiply). The script `averaging_search.py` checks Proposition 3 in exact arithmetic for $3 \leq d \leq 60$, $d + 3 \leq N \leq 400$. It prints the single pair $(6, 3)$.

3 The case $N = d + 3$: reduction to hard components

Lemma 5 (PSD forcing). *Let C be a connected graph on vertex set $[m]$, and let $x_1, \dots, x_m$ be unit vectors with $\langle x_u, x_v \rangle \neq 0$ iff $uv \in E(C)$ ($u \neq v$). Let $\nu = m - \dim \text{span}(x_v)$. If C is a tree, then $\nu \leq 1$. If C has exactly one cycle, then $\nu \leq 2$.*

Proof. ν is the dimension of $Z = \{c \in \mathbb{R}^m : \sum_v c_v x_v = 0\}$. Call a set B *forcing* if every $c \in Z$ vanishing on B vanishes everywhere; then $c \mapsto c|_B$ is injective on Z , so $\nu \leq |B|$. *Forcing rule.* Let $c \in Z$ vanish on B , and let K be a component of $C - B$. Vectors in different components of $C - B$ are orthogonal. So the vectors $\sum_{v \in K} c_v x_v$ (one for each K) are pairwise orthogonal and add up to 0, hence each is 0. If $w \in B$ has exactly one neighbour u in K , then $0 = \langle x_w, \sum_{v \in K} c_v x_v \rangle = c_u g_{wu}$, so $c_u = 0$ and u may be added to B .

Tree: $B = \{\text{root}\}$. A vertex of B has exactly one neighbour (its child) in each subtree hanging below it, so its children are forced, and by induction every vertex is. *One cycle* $v_1 \dots v_\ell$ ($\ell \geq 3$) with trees attached: $B = \{v_1, v_2\}$. Removing B , the vertex v_ℓ lies in a component in which v_1 has no other neighbour. So v_ℓ is forced, then $v_{\ell-1}$ by v_ℓ , and so on around the cycle. The attached trees are forced as in the tree case. Hence $\nu \leq |B|$ in both cases. $\square$

Theorem 6 (Reduction for $N = d + 3$). *Let x be a saturated minimiser of A for $d + 3$ lines in $\mathbb{R}^d$ (C5, Prop. 1). Then $A \geq 3\pi/2$, unless J has a component C with $r_C \geq 4$, $\nu_C \geq 3$, a vertex of degree 3 (and maximum degree 3), and cyclomatic number $|E(C)| - m_C + 1 \geq 2$.*

Proof. $\sum_C \nu_C = N - \sum_C r_C \geq 3$. We show $A_C \geq \min(\nu_C, 3) \frac{\pi}{2}$ for every component that is not of the excluded type. Since $\min(a, 3) + \min(b, 3) \geq \min(a + b, 3)$, this gives $A \geq \frac{3}{2}\pi$ whenever no excluded component exists.

$\nu_C \leq 2$. The $m_C = r_C + \nu_C$ lines lie in $V_C \cong \mathbb{R}^{r_C}$. By C3 ($\nu_C = 1$) and C5 ($\nu_C = 2$), $A_C \geq \nu_C \frac{\pi}{2}$.

$r_C \leq 3$. If $r_C \leq 2$: C1 in $\mathbb{R}^{r_C}$ (or coincident lines when $r_C = 1$) gives $A_C \geq M(m_C, r_C) \frac{\pi}{2} \geq (m_C - r_C) \frac{\pi}{2} = \nu_C \frac{\pi}{2}$. If $r_C = 3$ and $\nu_C \geq 3$: any 6 of the $m_C \geq 6$ lines satisfy Theorem 2, and $A_C \geq \frac{3}{2}\pi$ by monotonicity.

The rest: $r_C \geq 4$ and $\nu_C \geq 3$. Degrees are ≤ 3 (block lemma, $k = 3$). If C had maximum degree ≤ 2 , it would be an edge, a path or a cycle. The C5 analysis then gives $\nu_C \leq 2$ (a path on ≥ 3 vertices is impossible; a triangle or cycle has $\nu = 2$). So C has a vertex of degree 3. By Lemma 5, $\nu_C \geq 3$ excludes trees and unicyclic graphs, so the cyclomatic number is ≥ 2 . This is exactly the excluded type. $\square$

Corollary 7. *The conjecture holds for $N = d + 3$ when $d \leq 3$, and for every configuration of $d + 3$ lines whose saturated minimiser has no excluded component. Since $\sum r_C \leq d$ and an excluded component needs $r_C \geq 4$ and $m_C \geq 7$, the first open case is 7 lines in $\mathbb{R}^4$ with J connected.*

For $d \leq 3$ this is not new: $d = 2$ is C1 and $d = 3$ is Theorem 2. The corollary settles no new infinite family.

4 Why the general case stays open

For the excluded components there is no analogue of the cycle lemma. It would have to be an inequality $A_C \geq \frac{3}{2}\pi$ for vector families whose non-orthogonality graph has cyclomatic number ≥ 2 and maximum degree 3. The script `scan7.py` looks at the first open case, 7 lines in $\mathbb{R}^4$. It lists the 41 connected graphs on 7 vertices with maximum degree ≤ 3 and cyclomatic number ≥ 2 . For each, it minimises A with the orthogonality pattern imposed (penalty method, 4 random starts).

In *every* pattern the infimum found is $1.5\pi = M(7, 4)\frac{\pi}{2}$. It is approached only as some prescribed non-zero inner products tend to 0, i.e. at the boundary of the pattern, where the block structure changes. So no interior local minimum with connected J appeared below $\frac{3}{2}\pi$. This is consistent with the conjecture, but it is evidence only. A proof would need a potential (like Φ in C5) for graphs with two independent cycles.

Remark 8 (What we could not reproduce). Our working notes mentioned a rank-4, 7-line configuration with $A \approx 1.702\pi$ as a witness that the hard components are genuinely different. The file describing it (`p1_c6_codex.md`) was not available when this cell was written, and we did not reconstruct it. We therefore make no claim about it.

Scripts

- `averaging_search.py`: Proposition 3, exact arithmetic, ≈ 3 s.
- `scan7.py`: the pattern scan for 7 lines in $\mathbb{R}^4$ (numerical, not a proof), a few minutes.
- `search7.py`: unconstrained local minimisation for 7 lines in $\mathbb{R}^4$ (150 random starts). Every run ends at 1.5π , with J split into components of sizes $(1, 2, 2, 2)$ or $(2, 2, 3)$.

References

- [1] F. Fodor, V. Vigh, T. Zarnócz, On the angle sum of lines, *Arch. Math.* 106 (2016) 91–100.
- [2] D. Bilyk, R. Matzke, On the Fejes Tóth problem about the sum of angles between lines, *Proc. Amer. Math. Soc.* 147 (2019), arXiv:1801.07837.